

2023 - Fall

Pre-Board NIAST

Python (P.e)

- 18) A random sample of 10 bags ~~that~~ had the following I-Q: 70, 120, 110, 101, 83, 88, 95, 98, 107, 100. Find a reasonable range in which most of the mean I-Q. values of samples of 10 bags ~~to~~ lie. (used = 5%)

Here

Sample size, $n = 10$

X	X ²
70	4900
120	14400
110	12100
101	10201
83	6889
88	7744
95	9025
98	9604
107	11449
100	10000
972	96312

$$\bar{X} = \frac{972}{n} = \frac{972}{10} = 97.2$$

$$S = \sqrt{\frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2} = \sqrt{\frac{96312}{10} - (97.2)^2} = 13.541$$

Confidence level, $1-\alpha = 1 - 0.05 = 0.95$

level of significance, $\alpha = 5\% = 0.05$

we have,

$$t(\frac{\alpha}{2}, n-1) = t(\frac{0.05}{2}, 9) = 2.262$$

The 95% confidence interval is given by

$$C.I. = \bar{x} \pm t(\frac{\alpha}{2}, n-1) \frac{s}{\sqrt{n-1}}$$

$$= 97.2 \pm 2.262 \times \frac{13.541}{\sqrt{10-1}}$$

$$= 97.2 \pm 10.21$$

$$\therefore \text{lower limit} = 97.2 - 10.21 = 86.99$$

$$\therefore \text{upper limit} = 97.2 + 10.21 = 107.41$$

Hence, the reasonable range is $(86.99, 107.41)$.

ms

2022-Fall / 2023-Fall (Pre-Board) NAST

4.b) A population consists of value 7, 6, 8, 4, 10. Prove that the sample mean of size 3 is unbiased estimator of the population.

^{sol} In population, there are 5 units. The population mean is
$$\mu = \frac{\sum X_i}{n} = \frac{7+6+8+4+10}{5} = 7.$$

We have, 3 units can be selected from 5 units in
 ${}^5C_3 = 10$ ways.

Now,

Sample No.	Observations	$\bar{X}_i$	$P(\bar{X}_i)$
1.	7, 6, 8	$\frac{21}{3}$	$\frac{1}{10}$
2.	7, 6, 4	$\frac{17}{3}$	$\frac{1}{10}$
3.	7, 6, 10	$\frac{23}{3}$	$\frac{1}{10}$
4.	7, 8, 4	$\frac{19}{3}$	$\frac{1}{10}$
5.	7, 8, 10	$\frac{25}{3}$	$\frac{1}{10}$
6.	7, 4, 10	$\frac{21}{3}$	$\frac{1}{10}$
7.	6, 8, 4	$\frac{18}{3}$	$\frac{1}{10}$
8.	6, 8, 10	$\frac{24}{3}$	$\frac{1}{10}$
9.	6, 4, 10	$\frac{20}{3}$	$\frac{1}{10}$
10.	8, 4, 10	$\frac{22}{3}$	$\frac{1}{10}$

Now,

$$E(\bar{X}) = \sum_{i=1}^{10} \bar{X}_i P(X_i)$$

$$= \frac{1}{10} \times \frac{21}{3} + \frac{1}{10} \times \frac{17}{3} + \frac{1}{10} \times \frac{23}{3} + \frac{1}{10} \times \frac{19}{3} + \frac{1}{10} \times \frac{25}{3} +$$

$$\frac{1}{10} \times \frac{21}{3} + \frac{1}{10} \times \frac{18}{3} + \frac{1}{10} \times \frac{24}{3} + \frac{1}{10} \times \frac{20}{3} + \frac{1}{10} \times \frac{22}{3}$$

$$= \frac{1}{10} \times \left(\frac{21}{3} + \frac{17}{3} + \frac{23}{3} + \frac{19}{3} + \frac{25}{3} + \frac{21}{3} + \frac{18}{3} + \frac{24}{3} + \frac{20}{3} + \frac{22}{3} \right)$$

$$= \frac{1}{10} \times \frac{210}{3}$$

$$\therefore E(\bar{X}) = 7 = \mu$$

This shows that the sample mean of size 3 is unbiased estimator of the population.

2023-Fall (Pre-Board)

4.6) A population consists of value 7, 6, 8, 4, 10. Prove that sample mean of size 3 is unbiased of the population mean.

Sol In population, there are 5 units. The population mean is given by

$$\mu = \frac{\sum x}{n} = \frac{7+6+8+4+10}{5} = 7$$

The population ~~consists~~ of ~~53~~ ^{units} ~~which~~ which can be selected in 5C_3 ways = 10 ways.

Now,

Sample No.	Observations	$\bar{X}_i$	$P(\bar{X}_i)$
1.	7, 6, 8	$21/3$	$1/10$
2.	7, 6, 4	$17/3$	$1/10$
3.	7, 6, 10	$23/3$	$1/10$
4.	7, 8, 4	$19/3$	$1/10$
5.	7, 8, 10	$25/3$	$1/10$
6.	7, 4, 10	$21/3$	$1/10$
7.	6, 8, 4	$18/3$	$1/10$
8.	6, 8, 10	$24/3$	$1/10$
9.	6, 4, 10	$20/3$	$1/10$
10.	8, 4, 10	$22/3$	$1/10$

Now,

$$E(x) = \sum_{i=1}^{10} x_i \cdot P(\bar{x}_i)$$

$$= \frac{1}{10} \left(\frac{21}{3} + \frac{17}{3} + \frac{23}{3} + \frac{19}{3} + \frac{25}{3} + \frac{21}{3} + \frac{18}{3} + \frac{24}{3} + \frac{20}{3} + \frac{22}{3} \right)$$

$$\therefore E(x) = 7 = \mu$$

This shows that the sample mean of size 3 is unbiased of the population mean.

2022-Fall

5. a) A random sample of 8 envelopes is taken from a letter box of post office and their weights in grams are found to be 12.1, 11.9, 12.3, 11.9, 12.1, 12.4, 12.1 and 11.5. Find the 99% fiducial limits for the mean weight of envelopes received at the post office.

Given,

Sample size, $n = 8$

Samples: 12.1, 11.9, 12.3, 11.9, 12.1, 12.4, 12.1, 11.5

X	X^2
12.1	146.41
11.9	141.61
12.3	151.29
11.9	141.61
12.1	146.41
12.4	153.76
12.1	146.41
11.5	132.25
96.3	1159.75

$$\bar{X} = \frac{\sum X}{n} = \frac{96.3}{8} = ~~96.3~~ 12.0375$$

$$s = \sqrt{\frac{\sum X^2}{n} - \left(\frac{\sum X}{n}\right)^2} = \sqrt{\frac{1159.75}{8} - \left(\frac{96.3}{8}\right)^2}$$

$$= 0.26$$

confidence level, $1-\alpha = 99\%$

level of significance, $\alpha = 1\%$

Now,

$$t_{\left(\frac{\alpha}{2}, n-1\right)} = t_{\left(\frac{0.01}{2}, 7\right)} = 1.895$$

The 99% fiducial limits for the mean weight of envelopes received at the post office is given by

$$C.I. = \bar{x} \pm t_{\left(\frac{\alpha}{2}, n-1\right)} \cdot \frac{s}{\sqrt{n-1}}$$

$$= 12.0375 \pm 1.895 \times \frac{0.26}{\sqrt{8}}$$

$$= 12.0375 \pm 0.186$$

$$\therefore \text{lower limit} = 12.0375 - 0.186 = 11.8515 \text{ Ans}$$

$$\therefore \text{upper limit} = 12.0375 + 0.186 = 12.2235 \text{ Ans}$$

$$= 12.0375 \pm 0.186$$

$$\therefore \text{lower limit} = 12.0375 - 0.186 = 11.8515 \text{ Ans}$$

$$\therefore \text{upper limit} = 12.0375 + 0.186 = 12.2235 \text{ Ans}$$

2023-Spring / 2021-Spring

5. a) A random sample of 100 articles selected from a batch of 2000 particles shows that the average diameter of the article is 0.354 and standard deviation is 0.048. Find the 95% confidence limits for the average diameter of this batch of 2000 articles.

Sol

Total population, $N = 2000$

Sample selected, $n = 100$

level of significance, $\alpha = 5\%$, $\alpha = 0.05$

Confidence level, $1 - \alpha = 95\%$

Sample mean, $\bar{x} = 0.354$

Sample S.D, $s = 0.048$

we have,

$$Z_{\alpha/2} = \frac{Z_{0.05}}{2} = 1.96$$

$$\frac{0.5 - 0.05}{2} = 0.475$$

The 95% confidence limits for the average diameter is given by

$$C.I. = \bar{x} \pm Z_{\alpha/2} \frac{s}{\sqrt{n-1}} \sqrt{\frac{N-n}{N-1}}$$

$$= 0.354 \pm 1.96 \times \frac{0.048}{\sqrt{100-1}} \times \sqrt{\frac{2000-100}{2000-1}}$$

$$= 0.354 \pm 0.00921$$

∴

$$\therefore \text{lower limit} = 0.354 - 0.00921 = 0.34479$$

$$\therefore \text{upper limit} = 0.354 + 0.00921 = 0.36321$$

Ans

m

2021-Fall

5.a) A random sample of size 64 has been drawn from a population with standard deviation 20. The mean of the sample is 80. Calculate 95% confidence limits for the population mean. How does the width of the Confidence interval change if sample size is 256 instead?

SoP Given,

Sample size, $n = 64$

Sample S.D, $s = 20$

sample mean, $\bar{x} = 80$

confidence level, $1 - \alpha = 95\%$

level of significance, $\alpha = 5\% = 0.05$

we know,

$$Z_{\alpha/2} = Z_{\frac{0.05}{2}} = 1.96$$

$$\frac{0.5 - 0.05}{2}$$

The 95% confidence limits for the population mean is given by

$$= 0.475$$

(1.)

$$C.I. = \bar{x} \pm Z_{\alpha/2} \frac{s}{\sqrt{n}}$$

$$= 80 \pm 1.96 \times \frac{20}{\sqrt{64}}$$

$$\therefore C.I. = 80 \pm 4.9$$

$$\therefore \text{lower limit} = 80 - 4.9 = 75.1$$

$$\therefore \text{upper limit} = 80 + 4.9 = 84.9$$

when $n=256$,

$$C.I. = 80 \pm 1.96 \times \frac{20}{\sqrt{256}}$$

$$= 80 \pm 2.45$$

$\therefore$

$$\therefore \text{lower limit} = 80 - 2.45 = 77.55$$

$$\therefore \text{upper limit} = 80 + 2.45 = 82.45$$

When sample size is 256, the width of the confidence interval changes minimized.

2020-Fall / 2006 Spring

5.a) Consider a population of 5 units with 4, 2, 10, 6 & 8. Show that the sample mean of size 2 is unbiased estimator of population mean.

solⁿ In population, there are 5 units 4, 2, 10, 6 & 8. The population mean is:

$$\mu = \frac{4+2+10+6+8}{5} = 6$$

The 2 units from 5 units can be selected in 5C_2 ways = 10 ways.

Now,

Sample no.	Observations	$\bar{x}_i$	$P(\bar{x}_i)$
1.	4, 2	$6/2 = 3$	$1/10$
2.	4, 10	$14/2 = 7$	$1/10$
3.	4, 6	$10/2 = 5$	$1/10$
4.	4, 8	$12/2 = 6$	$1/10$
5.	2, 10	$12/2 = 6$	$1/10$
6.	2, 6	$8/2 = 4$	$1/10$
7.	2, 8	$10/2 = 5$	$1/10$
8.	10, 6	$16/2 = 8$	$1/10$
9.	10, 8	$18/2 = 9$	$1/10$
10.	6, 8	$14/2 = 7$	$1/10$

$$E(\bar{x}) = \sum \bar{x}_i P(\bar{x}_i) = \frac{1}{10} (3+7+5+6+6+4+5+8+9+7) \\ = \frac{1}{10} \times 60 = 6$$

Thus, the sample mean of size 2 is unbiased estimator of population mean.

2007-Fall

14. A population consists of value 1, 2, 3, 6, 8. Prove that the sample mean of size 3 is unbiased estimator of population mean.

99 In population, there are 5 units such as 1, 2, 3, 6, 8 and the population mean is given by

$$\mu = \frac{1+2+3+6+8}{5} = 4$$

These 3 units can be selected from 5 units in 5C_3 ways = 10 ways,

Now,

Sample no.	observations	$\bar{x}_i$	$P(\bar{x}_i)$
1.	1, 2, 3	$\frac{6}{3}$	$\frac{1}{10}$
2.	1, 2, 6	$\frac{9}{3}$	$\frac{1}{10}$
3.	1, 2, 8	$\frac{11}{3}$	$\frac{1}{10}$
4.	1, 3, 6	$\frac{10}{3}$	$\frac{1}{10}$
5.	1, 3, 8	$\frac{12}{3}$	$\frac{1}{10}$
6.	1, 6, 8	$\frac{15}{3}$	$\frac{1}{10}$
7.	2, 3, 6	$\frac{11}{3}$	$\frac{1}{10}$
8.	2, 3, 8	$\frac{13}{3}$	$\frac{1}{10}$
9.	2, 6, 8	$\frac{16}{3}$	$\frac{1}{10}$
10.	3, 6, 8	$\frac{17}{3}$	$\frac{1}{10}$
Total:		40	1

Now,

$$E(\bar{x}) = \sum (\bar{x}_i) P(\bar{x}_i) = 40 \times \frac{1}{10} = 4 = \mu$$

This shows that the sample mean of size 3 is unbiased estimator of population mean.